Fast exact overlap minimization for three-dimensional irregular strip packing of polyhedra

Ignacio Araya[a,*]

[a]*Escuela de Ingeniería Informática, Pontificia Universidad Católica de Valparaíso, Chile*

Abstract

We address the three-dimensional irregular strip packing problem: polyhedra, possibly non-convex, must be placed by translation in a container with a fixed rectangular base so that the height is minimized. We propose an overlap-minimization heuristic that works with the exact geometry of the pieces, represented as unions of convex parts. Its new element is an exact axis move. The overlap of two parts is measured by the penetration depth of their relative position in their no-fit polytope; along any axis this function is zero outside an interval and concave inside it, so the best position of a piece along an axis is found by evaluating a few breakpoints only. The rest of the method adapts a few known components from two-dimensional nesting: a greedy construction, a strategic oscillation of the container height with guided local search weights, a local improvement and restarts. On the benchmarks of polyhedral pieces, the method improves the best known heights of 11 of 13 instances within two minutes, reaches in 10 seconds the quality that the previous exact local search attains in 250 seconds (time normalized to our hardware), and matches 24 of the 27 known optima of small instances. An ablation study shows that the speed of the move, the guided local search weights and the effort spent in each separation are the decisive elements.

Keywords: Cutting, Packing, 3D irregular packing, No-fit polytope, Guided local search, Linear programming

*Corresponding author

Email address: ignacio.araya@pucv.cl (Ignacio Araya)

1. Introduction

Three-dimensional irregular packing arises in additive manufacturing (packing parts in the build volume of a printer), in the loading of non-rectangular goods, and in the design of compact assemblies. In its open-dimension variant, the *3D irregular strip packing problem*, a set of polyhedra must be placed without overlap inside a container with a fixed $W \times D$ base, and the height of the packing is to be minimized. We consider the translational version: pieces keep their orientation.

The literature on this problem is small compared with its two-dimensional counterpart, and it is split by the representation of the geometry. Exact approaches use the polyhedra directly: phi-functions and nonlinear programming (Stoyan et al., 2004, 2005; Romanova et al., 2018), the 3DNest local search of Egeblad et al. (2009), which minimizes the exact overlap volume along axis-parallel translations, constructive heuristics such as HAPE3D (Liu et al., 2015), and, recently, mixed-integer models based on no-fit polyhedra that solve small convex instances to optimality (Tollenaere et al., 2026). Approximate approaches discretize the pieces into voxels (Lamas-Fernandez et al., 2023); voxels make overlap tests trivial and scale well, but they overestimate the pieces and the quality of the solutions depends on the resolution.

In two dimensions, the most successful methods for irregular strip packing are overlap-minimization heuristics: the height (length) is decreased, the pieces are allowed to overlap, and a local search moves them until the overlap vanishes (Egeblad et al., 2007; Umetani et al., 2009; Imamichi et al., 2009; Elkeran, 2013; Gardeyn et al., 2025). The key ingredients are a cheap and informative overlap measure, moves that can be optimized exactly or quickly, and guided local search (GLS) weights to escape local minima. This paper brings these ingredients to three-dimensional polyhedra with an exact representation. Most components of the method are adaptations of known ideas, whose origin is given in Section 3. The new element is the overlap measure and the move built on it: the penetration depth in the no-fit polytopes (NFPs) of convex parts which, unlike the overlap volume used by 3DNest, can be minimized exactly along an axis by evaluating a few breakpoints.

Contributions.

1. A fast exact overlap-minimization move for 3D polyhedra represented as unions of convex parts. It brings the exact axis moves of Umetani et al. (2009), defined for 2D polygons with directional penetration

depths, to 3D and non-convex pieces: the overlap is measured by the penetration depth in the NFP of each pair of convex parts, which along an axis is concave on an interval and zero outside it, so the optimal position of a piece along an axis is attained at an interval endpoint (Proposition 1). With an implementation that evaluates each term only at the breakpoints inside its interval, the move is much cheaper than the exact overlap volume of Egeblad et al. (2009), and this speed is what makes exact geometry practical in a 3D metaheuristic.

2. A method built around this move with few components, adapted from known methods (a bottom-left construction with gravity, a strategic oscillation of the height, multiplicative GLS weights (Gardeyn et al., 2025), a local improvement and restarts), and an ablation study showing which elements matter: the GLS weights, the speed of the move and the effort spent in each separation are decisive, the local improvement is needed to reach the optima, and the perturbation of the restarts and the tuned construction criterion have no significant effect.
3. New best known solutions on the Merged instances of Egeblad et al. (2009) and on classical instances of Stoyan et al. (2004, 2005), with a time-matched comparison against the exact method 3DNest, and a comparison with the optimal values and bounds of the mixed-integer models of Tollenaere et al. (2026).
4. An exact convex decomposition of the non-convex (prismatic) pieces of Ikonen et al. (1997) used in the Merged instances.

Section 2 defines the problem and the geometric tools. Section 3 describes the algorithm, and Section 4 the experiments.

2. Problem and geometry

2.1. Problem

We are given a container base of width $W > 0$ and depth $D > 0$, of unbounded height, and n pieces $P_1, \ldots, P_n \subset \mathbb{R}^3$ (polyhedra, with multiplicities). Each piece is a union of convex polytopes, $P_i = \bigcup_{a \in \mathcal{A}_i} A_a$, expressed in a local frame where the minimum of its bounding box is the origin; w_i, d_i, h_i denote its bounding-box dimensions. A solution is a vector of translations $t_i = (x_i, y_i, z_i) \in \mathbb{R}^3$, $i = 1, \ldots, n$ (continuous positions; rotations are not

allowed), such that

$$0 \le x_i \le W - w_i, \quad 0 \le y_i \le D - d_i, \quad z_i \ge 0, \qquad i = 1, \dots, n, \tag{1}$$

$$\operatorname{int}(P_i + t_i) \cap \operatorname{int}(P_j + t_j) = \emptyset, \qquad i < j, \tag{2}$$

where $\operatorname{int}(\cdot)$ denotes the interior of a set (pieces may touch but not overlap), and the objective is to minimize $H = \max_i(z_i + h_i)$. A trivial lower bound is $H_0 = \max\{\max_i h_i,\ \sum_i \operatorname{vol}(P_i)/(WD)\}$.

2.2. No-fit polytopes and penetration depth

For two convex parts A and B placed at t_A and t_B, their interiors intersect if and only if the relative position $r = t_A - t_B$ lies in the interior of the no-fit polytope

$$N_{AB} = B \oplus (-A) = \operatorname{conv}\{b - a :\ a \in V(A),\ b \in V(B)\}, \tag{3}$$

where $V(\cdot)$ is the vertex set (Bennell and Oliveira, 2008). We compute N_{AB} once per pair of parts as the convex hull of the vertex differences and store it in H-representation, $N_{AB} = \{r :\ n_f \cdot r \le d_f,\ f \in F_{AB}\}$, with unit normals $\|n_f\| = 1$.

Definition 1. *The* penetration *of A and B at relative position r is* $\delta_{AB}(r) = \max\{0,\ \min_{f \in F_{AB}}(d_f - n_f \cdot r)\}$.

For r inside N_{AB}, $\delta_{AB}(r)$ is the Euclidean distance from r to the boundary of N_{AB}, i.e., the length of the shortest translation that separates A and B, known as the translational penetration depth (Cameron and Culley, 1986; Agarwal et al., 2000). It is zero if and only if the parts do not overlap. For two pieces we use $\delta_{ij}(t_i - t_j) = \sum_{a \in \mathcal{A}_i} \sum_{b \in \mathcal{A}_j} \delta_{ab}(t_i - t_j)$, where only pairs of parts whose bounding boxes intersect are evaluated. Figure 1 illustrates both definitions in two dimensions.

2.3. Exact minimization along an axis

Let all pieces but i be fixed, and let i move along axis e_k, $t_i(s) = t_i + (s - t_{ik})e_k$, with $s \in [0, U_k]$ (the container limits, using the current height H for the vertical axis). The weighted overlap of i is

$$\phi(s) = \sum_{j \ne i} w_{ij}\, \delta_{ij}\big(t_i(s) - t_j\big), \tag{4}$$

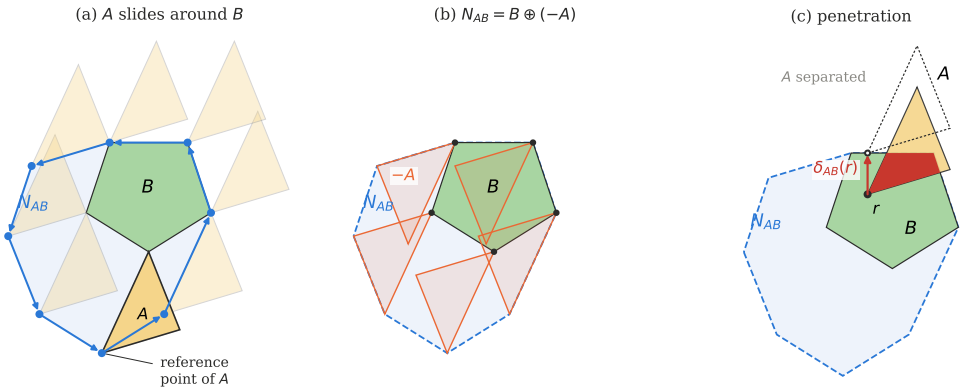


Figure 1: No-fit polytope and penetration in 2D. (a) When A slides around B keeping contact, its reference point (the minimum of its bounding box) traces the boundary of N_{AB}; the shaded copies of A touch B at the vertices of N_{AB}. (b) Equivalently, $N_{AB} = B \oplus (-A)$ is the convex hull of the copies of $-A$ placed at the vertices of B. (c) If r lies inside N_{AB} the parts overlap (red), and $\delta_{AB}(r)$ is the distance from r to the nearest facet: the shortest translation of A that separates the parts (dotted).

where the weights $w_{ij} \geq 1$ (initially 1) are the penalties of guided local search (GLS) (Voudouris and Tsang, 1999): they are increased for the pairs that remain overlapping at local minima, so that the search focuses on them (Section 3.4).

Proposition 1. *Each term $\delta_{ab}(t_i(s) - t_j)$ is zero outside an interval (L_{ab}, U_{ab}) and concave inside it. Hence ϕ is concave on every interval between consecutive points of $\mathcal{B} = \{0, U_k\} \cup \{L_{ab}, U_{ab}\}$, and a minimizer of ϕ on $[0, U_k]$ is attained at a point of $\mathcal{B}$.*

Proof. Along the line, $d_f - n_f \cdot r(s) = c_f - n_{fk} s$ is affine. The set where all of them are positive is the open interval (L_{ab}, U_{ab}) with $L_{ab} = \max_{n_{fk}<0} c_f/n_{fk}$ and $U_{ab} = \min_{n_{fk}>0} c_f/n_{fk}$ (facets with $n_{fk} = 0$ only decide whether the interval is empty). Inside it, δ_{ab} is a minimum of affine functions, hence concave; outside it is zero. Between consecutive points of $\mathcal{B}$ every term is either concave or identically zero, so ϕ is concave there and attains its minimum at an endpoint. □

Proposition 1 is the counterpart, for the Euclidean penetration depth of convex parts in 3D, of the piecewise linear directional penetration depths of Umetani et al. (2009) in 2D, which require a trapezoidal decomposition of each NFP and have no practical equivalent for 3D NFPs. With pieces

decomposed into convex parts, every NFP is convex and each term reduces to a minimum of affine functions. Figure 2 illustrates the proposition in two dimensions.

The move evaluates ϕ at the points of $\mathcal{B}$ only. With the interval of each term known, a term is evaluated only at the candidates inside it: sorting the candidates, each term contributes to a contiguous range found by binary search. Facets are stored as (c_f, n_{fk}) in contiguous arrays, so evaluating a term costs one fused multiply-add per facet. Ties are broken towards the smaller coordinate (gravity and walls).

3. Algorithm

Algorithm 1 summarizes the method. A greedy construction, followed by a local improvement, gives a first feasible solution S of height H. The search then tries to fit the pieces in a slightly lower container: the positions of S are *clamped* to a target height $h < H$: Clamp(S, h) moves every piece that exceeds h down until its top is at h, regardless of the other pieces. This is where overlaps appear, and MinimizeOverlap tries to remove them. If it succeeds, the layout is feasible and fits in height h; it is improved locally and becomes the new best solution, and the target is lowered again; otherwise the target is raised slightly and, when it reaches H, the search restarts from the best solution.

In Algorithm 1, line 1 builds the first solution (Sections 3.1 and 3.2) and line 2 creates the first overlapping layout. Each iteration of the loop is a separation attempt at height h (Section 3.4). On success (lines 5–6) the separated layout is improved, becomes the new best solution, and the target is lowered again. On failure (lines 8–10) the target is raised by a quarter of the step and the search continues from the current overlapping layout with the current weights; once the target is back at the best height, the search restarts from the best solution (Section 3.5).

3.1. Greedy construction

Pieces are placed one at a time on top of the current load (Algorithm 2). For each type of piece still to be placed, $g \times g$ starting points ($g = 4$) are taken on a regular grid of the base, at the current height H_{cur} of the load, and from each of them the piece descends (Descend, lines 9–18). The descent is a sequence of classical linear programs (LPs) in the three translation variables $u = (x, y, z)$ of the piece. At the current point t, for every placed part b

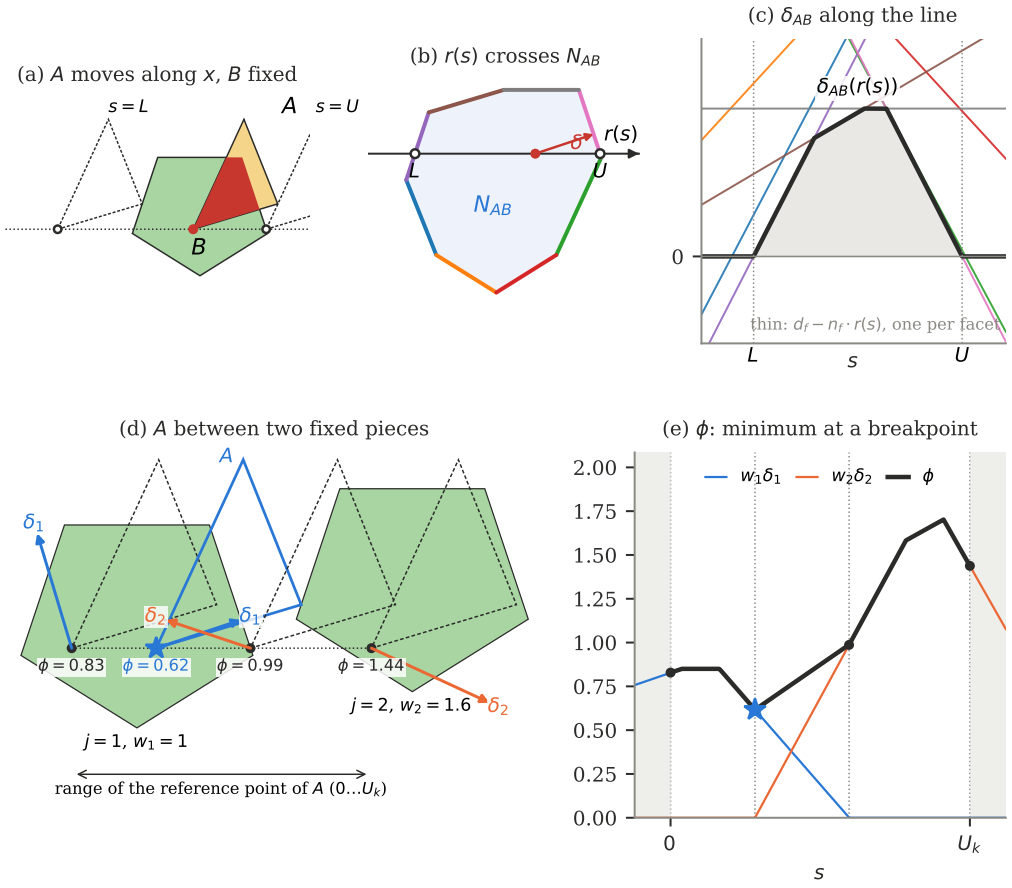


Figure 2: Proposition 1 in 2D, with the parts of Figure 1. (a) Part A moves along x with B fixed: it touches B at $s = L$ and $s = U$ (dashed) and overlaps it in between (red). (b) In the space of relative positions, $r(s)$ moves along a line that enters N_{AB} at L and leaves it at U. (c) For each facet (same colors as in (b)), $d_f - n_f \cdot r(s)$ is affine in s; the penetration is the lower envelope of these lines clipped at zero: zero outside (L, U) and concave inside. (d)–(e) A moves between two fixed pieces within the container range $[0, U_k]$. In (d), A is drawn at each breakpoint of (e) (0, U_k and the L, U of each term; dots: its reference point), with arrows for its penetration with each piece it overlaps (δ_1, δ_2) and the resulting $\phi = w_1\delta_1 + w_2\delta_2$ (weights $w_1 = 1$, $w_2 = 1.6$): the two central positions have similar penetrations but different ϕ. Between consecutive breakpoints the weighted overlap ϕ is a sum of concave functions, so its minimum (star, solid outline in (d)) is at a breakpoint, and only these points are evaluated.

Algorithm 1: Overview

```
1  S ← Improve(Greedy()); H ← H(S);
2  h ← (1 − ρ)H; T ← Clamp(S, h); w ← 1;            // overlaps appear
3  while time remains do
4      if MinimizeOverlap(T, h, w) then              // T is feasible
5          S ← Improve(T); H ← H(S);   // new best solution, H ≤ h
6          h ← (1 − ρ)H; T ← Clamp(S, h); w ← 1;
7      else
8          h ← h + ρH/4; T ← Clamp(T, h);        // keeps the overlaps
9          if h ≥ H then
10             h ← (1 − ρ)H; T ← Clamp(S, h); move a random piece
                of T; w ← 1;                                  // restart
11 return S;
```

that the part a of the piece may meet within a trust region (a box of half side Δ, the largest dimension of the piece), one facet $n_f \cdot r \geq d_f$ of N_{ab} satisfied at t is selected: the one that allows the longest step in the descent direction (line 13). Since N_{ab} is convex, the half-space beyond any of its facets contains no overlapping position, so the selected facets define a convex region of positions where the piece does not overlap the placed parts. The LP of line 14 is

$$\begin{aligned} \min_{u} \quad & c \cdot u \\ \text{s.t.} \quad & n_f \cdot (u - t_b) \geq d_f \quad \text{for every selected facet } f \text{ of } N_{ab}, \\ & \max(\ell,\, t - \Delta) \leq u \leq \min(\bar{u},\, t + \Delta), \quad u_z \leq t_z, \end{aligned} \tag{5}$$

where t_b is the position of the piece of b and $[\ell, \bar{u}]$ are the positions that keep the piece inside the container. The objective $c = (\pm\sigma, \pm\sigma, 1)$ moves the piece down and, with a small weight σ, towards the walls in x and y nearest to the starting point (the signs are fixed at the start, so the objective is linear). The current point t is feasible, and every feasible point is a position without overlap, so each LP can only improve the position. With three variables and a few dozen constraints, the LP is solved by a small primal active-set method (a simplex method in the space of u) that starts at t and keeps every iterate feasible; no external solver is needed. The facets are re-selected at the new point and the procedure stops when the objective

no longer decreases. Figure 4 shows the LPs of one descent. DESCEND combines two classical ideas: bottom-left placement rules for irregular pieces (Jakobs, 1996), here with priority to gravity, as the minimum potential energy principle of HAPE3D (Liu et al., 2015), and the compaction of a layout by linear programs over NFP facets (Li and Milenkovic, 1995; Gomes and Oliveira, 2006), here applied to a single piece.

Among all piece types and resulting positions, the greedy places the pair (k, t) that maximizes (line 6)

$$val(k,t) = \alpha \log \frac{V_k}{V_{\max}} - \beta \frac{z}{H_0} - \gamma \frac{\max(0,\, z + h_k - H_{\text{cur}})}{H_0}, \tag{6}$$

where V_k is the volume of the piece, $V_{\max}$ the largest volume, z the height of the position and H_{cur} the current height of the load. The first term favors large pieces, as the classical orderings by decreasing volume, but on a logarithmic scale, so that a smaller piece is preferred when it fits clearly lower. The second term favors low positions, i.e., filling the gaps of the load before building on top of it, and the third penalizes positions that raise the height of the load. Normalizing by the lower bound H_0 makes the parameters independent of the scale of the instance. For a given piece the score decreases with z, so the lowest of its positions is always preferred; the score thus decides which piece is placed next: a large piece if it fits in a low gap, and a smaller one if the large pieces would raise the height. The parameters ($\alpha = 1$, $\beta = 5.53$, $\gamma = 0.77$, $\sigma = 0.065$) were tuned with Optuna on 16 training instances, for an earlier version of the method that relied more on the construction. Figure 3 illustrates the construction in two dimensions.

3.2. Local improvement

IMPROVE is applied to the feasible layouts: after the construction and after each successful separation. It is a compaction of the layout in the sense of Li and Milenkovic (1995), performed one piece at a time with DESCEND. Taking the pieces in increasing order of z, every piece descends again with DESCEND, with the other pieces fixed, and each piece that defines the height is moved to the lowest of the positions generated as in the greedy construction, if it is lower. The pass is repeated while the height decreases (at most 50 passes). Without this step the height of the best solution could only decrease in the discrete steps of the oscillation (Section 3.3), since a separated layout fits in the target height but its pieces are not pushed down; with it,

Algorithm 2: Greedy construction

```
function GREEDY()
    L ← ∅ (placed pieces); H_cur ← 0;
    while some piece is not placed do
        foreach type k with pieces left and each of g × g grid points (x, y) of the base do
            t ← DESCEND(k, (x, y, H_cur), L);
            val ← val(k, t) of Eq. (6);
        place the pair (k, t) of largest val in L;
        H_cur ← max(H_cur, z + h_k);
    return L;
function DESCEND(k, t, L)
    c ← (±σ, ±σ, 1);                    // down and towards the nearest walls
    repeat
        foreach part a of k and placed part b of L that may meet a in the trust region do
            select the facet f of N_ab satisfied at t that allows the longest step along −c;
        t' ← optimal solution of the LP (5) built at t;
        if c · t' ≥ c · t then break;
        t ← t';
    until no improvement;
    return t;
```

the pieces settle onto each other and, for instance, the optima of the small instances are reached exactly (Section 4.4).

3.3. Strategic oscillation

After a successful separation the target height is decreased by $\rho = 1\%$ of the best height and the best solution is clamped to it,

$$\textsc{Clamp}(T, h): \quad z_i \leftarrow \min\{z_i,\ h - h_i\} \text{ (and } x_i, y_i \text{ projected onto the container)}, \qquad i = 1, \dots, n, \tag{7}$$

which creates overlaps among the pieces near the top. After a failure the target is increased by $\rho/4$ and the search continues from the overlapping layout,

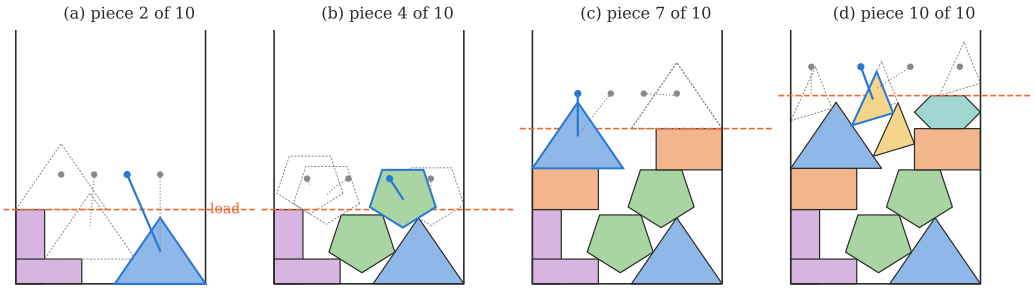


Figure 3: Greedy construction on a 2D example (10 pieces, one of them a union of two convex parts), computed with a 2D version of the method. In each step, a piece starts from $g = 4$ points on top of the current load (dots) and descends by a sequence of LPs over NFP facets, towards the floor and the nearest wall; dotted outlines: final positions of the other starts; blue: descent and position chosen by the greedy criterion.

with the GLS weights of the previous attempt. This scheme of shrinking the container, allowing overlap and separating the pieces comes from the 2D methods of Egeblad et al. (2007), Umetani et al. (2009) and Imamichi et al. (2009), and was used in 3D by Egeblad et al. (2009); the clamp is the counterpart of moving the pieces that exceed the new strip length inside it.

3.4. Overlap minimization

MinimizeOverlap performs passes over the overlapping pieces in random order; for each piece the three axes are tried in random order with the exact move of Proposition 1. Moving one piece at a time along the axes to the position of minimum overlap is the neighborhood of Egeblad et al. (2007, 2009), who minimize the overlap area or volume; Umetani et al. (2009) and Imamichi et al. (2009) measure the overlap of 2D polygons by the penetration depth in the NFP, which we use for 3D unions of convex parts. The pairwise penetrations are kept in a matrix and only the row of a moved piece is recomputed. When a pass does not decrease (4), the weights of guided local search (Voudouris and Tsang, 1999), used for nesting by Egeblad et al. (2007) and Umetani et al. (2009), are updated multiplicatively as in Gardeyn et al. (2025): the weight of every overlapping pair is multiplied by $1.2 + 0.8\,\delta_{ij}/\max\delta$, and the other weights decay by a factor 0.95 (minimum 1). The procedure stops when no overlap remains (then verified with a separating-axis test) or after 500 passes. Weights persist along the oscillation

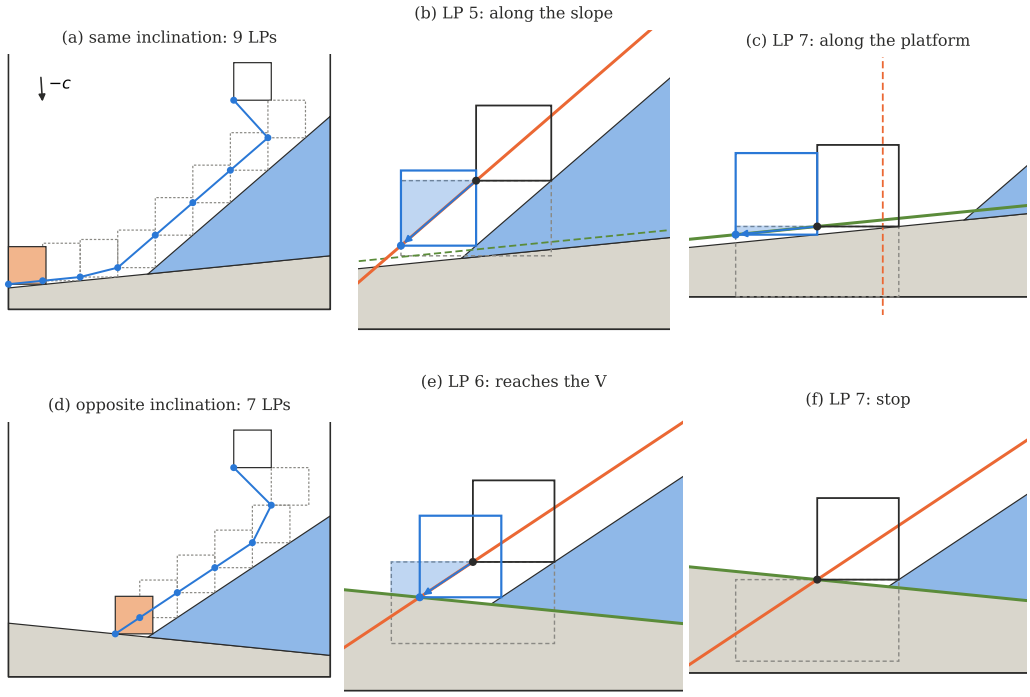


Figure 4: DESCEND step by step on a 2D example: a square falls onto a slope that ends on a platform whose top is slightly inclined, (a)–(c) in the same direction as the slope, (d)–(f) in the opposite direction. (a), (d) The whole descent: each dot is the reference point of the square (the corner of its bounding box) after one LP, and the dotted boxes are the trust regions. In free space the optimum of the LP is a corner of the trust region, so the square moves down and towards the nearest wall; once it touches the slope it slides along it. (b), (c), (e), (f) Single LPs (Algorithm 2, line 14) in the space of the reference point: t is the current point, the dashed box the trust region (the point may not go up), each line the facet selected for a placed part (orange: slope; green: platform; solid if active at the optimum), the shaded region the feasible region, and the arrow the move to the optimum along $-c$. (b) The square slides along the slope; the facet of the platform is already selected but not active yet. (c) On the platform, the facet of the platform is active and the square keeps sliding, down the slight incline, to the wall. (e) With the opposite inclination, the square reaches the V between the slope and the platform. (f) Both facets are active and no direction decreases $c \cdot t$: the descent stops. Gravity has priority over the wall term, so the piece slides down any incline.

and are reset after a successful separation or a restart.

In Algorithm 3, line 2 computes the penetrations of all pairs, and each iteration of the loop is a pass. Only the pieces involved in an overlapping pair are moved (line 7), each along the three axes with the exact move of Proposition 1 (line 9), in which the weighted overlap ϕ of Eq. (4) is evaluated at the breakpoints and the best one is taken; since the current position is a candidate, a move never increases ϕ. After moving a piece, only its row of the penetration matrix is recomputed (line 11). A pass without gain (line 12) is a local minimum of the weighted overlap; the weights of the pairs that still overlap are then increased in proportion to their penetration, and those of the other pairs decay, which changes ϕ and lets the next pass move the pieces out of the local minimum.

Algorithm 3: Overlap minimization

```
1  function MinimizeOverlap(T, h, w)
2      P_ij ← δ_ij(t_i − t_j) for all pairs;                    // penetration matrix
3      for pass = 1, ..., 500 do
4          O ← {(i, j) : P_ij > 0};
5          if O = ∅ then return true if the separating-axis test
             confirms it;
6          gain ← 0;
7          foreach piece i of a pair of O, in random order do
8              foreach axis k ∈ {x, y, z} in random order do
9                  s* ← arg min_{s∈B} φ(s) over the breakpoints B of
                     Proposition 1, with the container
                     [0, W − w_i] × [0, D − d_i] × [0, h − h_i];
10                 gain ← gain + φ(t_ik) − φ(s*); t_ik ← s*;
11             recompute row i of P;
12         if gain = 0 then                                     // local minimum: GLS
13             w_ij ← max(1, 0.95 w_ij) for (i, j) ∉ O;
               w_ij ← w_ij (1.2 + 0.8 P_ij / max_O P) for (i, j) ∈ O;
14     return false;
```

Figure 5 shows all the passes on the 2D example, from a clamped layout.

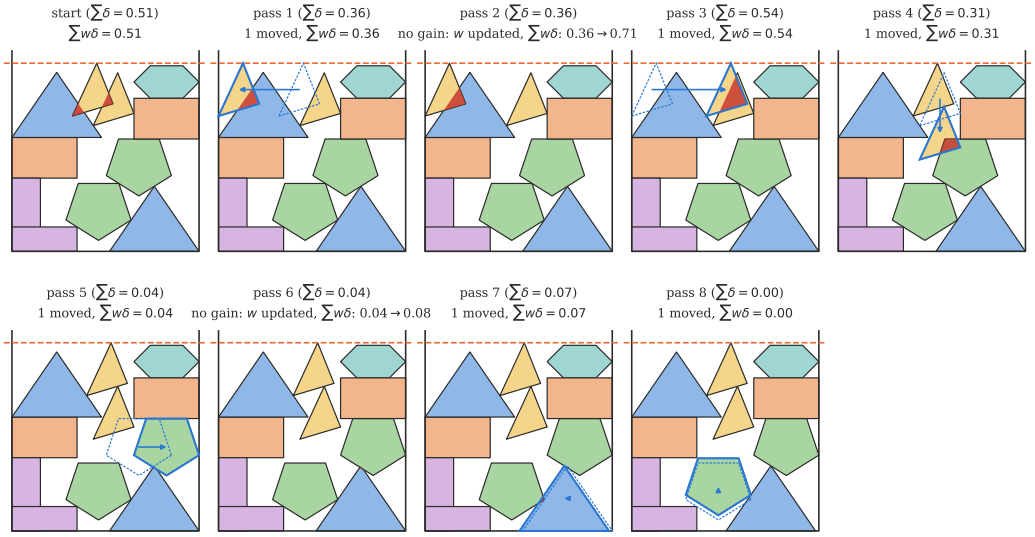


Figure 5: MinimizeOverlap on the 2D example: the greedy layout of Figure 3 is clamped to the target height $h = 0.9H$ (dashed line), and every pass is shown. In each pass, the pieces that overlap are moved along the axes; the pieces actually moved are outlined in blue (dashed: position before the pass), and overlaps are in red. Each title gives the unweighted overlap $\sum \delta$ and the weighted overlap $\sum w\delta$, which is the one minimized: no move increases it. Pass 2 does not reduce it (a local minimum), so the weight of the overlapping pair of triangles is increased from 1 to 2 and $\sum w\delta$ jumps from 0.36 to 0.71; the weights change the objective, not the layout, and the next pass starts from the same positions. Pass 3 then moves the small triangle onto the other one: $\sum w\delta$ decreases to 0.54, although $\sum \delta$ increases. The triangle is moved down in pass 4. The same happens with the pentagon and the triangle at the bottom in passes 6–8, and after pass 8 the layout is feasible.

3.5. Restarts

When the target reaches the best height, the search restarts from the best solution clamped to $(1-\rho)H$, with the weights reset, after moving one random piece (a kick) to a random position in the container. This is the perturbation step of iterated local search (Lourenço et al., 2003), which Imamichi et al. (2009) applied to nesting. On the larger instances the kick itself has no measurable effect (Table 2): the diversification comes from the restart, with reset weights and a new random order of the moves. The random positions are needed on small instances: with two pieces there is a single pair, so the weights have no effect, and the search keeps returning to the layout produced by the greedy order (Section 4.4).

4. Computational experiments

4.1. Setting

The method is implemented in C++ and needs no external solver. Experiments ran on two Intel Xeon E5-2620 v4 processors (2.1 GHz, 3.0 GHz turbo, 16 cores, 64 GB, Ubuntu 24.04, g++ 13.3), one thread per run, 14 runs in parallel. Each configuration was run with 10 seeds and a time limit of 120 s per instance. All reported solutions were verified with an independent checker that tests every pair of convex parts for a separating axis, with a tolerance of 10^{-6} on the penetration depth (the containers are 10 to 60 units wide). Table 1 lists the parameters of the method. The parameters of the greedy construction were tuned with Optuna on 16 training instances for an earlier version of the method and were not re-tuned: replacing the tuned criterion by the largest piece first changes the final heights by only +0.3% ($p = 0.22$, Table 2), so the final quality depends little on them; the weight update follows Gardeyn et al. (2025); the step ρ and the number of passes are analyzed in Table 2.

Instances. We use the 3D instances of the ESICUP repository (EURO Special Interest Group on Cutting and Packing, 2025):

- *Stoyan 2005*, Examples 1–3 (7, 12 and 25 convex pieces) (Stoyan et al., 2005). In the public data, one piece (polytope7) has faces that use only 9 of its 10 vertices; we use its convex hull (volume 192.5), which matches the volume reported by Egeblad et al. (2009).

Table 1: Parameters of the method.

Parameter	Value	Role	Origin
g	4	starting points per axis of the base	tuned
α, β, γ	1, 5.53, 0.77	greedy criterion (6)	tuned
σ	0.065	wall term of DESCEND	tuned
Δ	largest dimension of the piece	trust region of DESCEND	design
ρ	1% (raised by $\rho/4$)	step of the oscillation	Table 2
passes	500	limit of MINIMIZEOVERLAP	Table 2
GLS	$\times(1.2 + 0.8\,\delta_{ij}/\max\delta)$; decay 0.95	weight update at local minima	Gardeyn et al. (2025)
IMPROVE	at most 50 passes	local improvement	safeguard
kick	one random piece, random position	restart	Table 2

- *Stoyan 2004*, Examples 1–5 (20–50 non-convex pieces) (Stoyan et al., 2004), each defined as a union of convex polytopes, which we use directly as the decomposition.
- *Merged1–5* (Egeblad et al., 2009): i copies of 15 pieces, the 7 convex pieces of Stoyan 2005 and the 8 pieces of Ikonen et al. (1997) scaled by 4, in a container whose base is such that a layout with 50% utilization is a cube. Six Ikonen pieces are non-convex prisms; we decompose them exactly by triangulating the base polygon, merging triangles into convex polygons (Hertel and Mehlhorn, 1983), and extruding (a frame is split into four boxes; two pieces have polygons touching at a vertex). The volumes of the parts match those of the meshes.
- *Liu Experiment 1* (Liu et al., 2015): 36 pieces (8 tetrahedra, 8 cubes, 8 stars, 8 arrows and 4 rings) in a 20×20 container; the stars, arrows and rings are prisms and are decomposed exactly in the same way.

References. 3DNest (Egeblad et al., 2009) uses the same exact geometry; its average and maximum utilization over 10 runs of 600 s (one core of a 2.16 GHz Core Duo) are converted to heights. Lamas-Fernandez et al. (2023) report best and average heights of a voxel VNS; their voxelization overestimates the pieces (by 15% in volume for Merged1, as they note), so their heights are not directly comparable. For Stoyan 2004 the only exact reference is Stoyan et al. (2004).

4.2. Component analysis

Table 2 evaluates the components on the seven hardest instances (Stoyan 2004 Examples 1–5 and Stoyan 2005 Examples 2–3). Each row changes one element of the method:

- *Without Improve*: no local improvement after the construction and the separations (Section 3.2).
- $\sigma = 0$: DESCEND without the wall term (gravity only).
- *Greedy by volume*: the largest piece first, at its lowest position (α large, $\gamma = 0$), instead of the tuned criterion (6).
- *No kick*: the restarts do not move any piece.
- *Additive weights*: $w_{ij} \leftarrow w_{ij} + 1$ for the overlapping pairs at local minima.
- *Time limit of 24 s*: the same runs stopped at 24 s instead of 120 s. A fifth of the time is what the method would have if its moves were five times more expensive, e.g., with a costlier overlap measure, so this row estimates how much of the performance is due to the speed of the exact axis move of Proposition 1.
- *Sensitivity*: step ρ of the oscillation and maximum number of passes of MINIMIZEOVERLAP.

For each variant, the last column of the table gives the mean relative change of the height with respect to the proposed method over the 70 runs (7 instances, 10 seeds), positive when the heights are higher (worse), and the p-value of a Wilcoxon signed-rank test that pairs the runs by instance and seed.

The decisive elements are the GLS weights and the effort spent in each separation. Additive weights increase the heights by 1.4% ($p < 0.001$). With a fifth of the time (24 s), as with moves five times more expensive, the heights increase by 2.3% ($p < 0.001$) and most of the advantage over the published values is lost (the gap on Stoyan 2004 drops from -2.9% to -0.1%): the speed of the move is as important as its exactness. The number of passes of MINIMIZEOVERLAP also matters: with 100 passes the heights increase by 1.8% ($p < 0.001$), since separations that would succeed are abandoned, and with 2000 passes by 0.7% ($p = 0.008$), since the time is spent on hopeless

Table 2: Component analysis on the seven hardest instances (120 s, 10 seeds). Gap: mean gap of the height with respect to the best published value (Stoyan 2004 Examples 2–5 and Stoyan 2005 Examples 2–3); negative is better. Change: mean relative change of the height with respect to the proposed method over the 70 runs, i.e., over the seven instances including Stoyan 2004 Example 1, which has no published reference; positive means higher, i.e., worse (p: Wilcoxon signed-rank test).

	Gap		
Configuration	S2004 Ex2–5	S2005 Ex2–3	Change (p)
Proposed method	−2.9%	−2.0%	–
Without IMPROVE	−2.6%	−1.6%	+0.4% (0.10)
$\sigma = 0$ (no wall term in DESCEND)	−2.3%	−1.7%	+0.5% (0.19)
Greedy by volume	−2.9%	−1.4%	+0.3% (0.22)
No kick (restarts only)	−2.6%	−2.1%	+0.2% (0.29)
Additive GLS weights	−1.4%	−0.8%	+1.4% (<0.001)
Time limit 24 s (default 120 s)	−0.1%	−1.2%	+2.3% (<0.001)
$\rho = 0.5\%$ (default 1%)	−2.8%	−2.1%	+0.1% (0.81)
$\rho = 2\%$	−2.2%	−1.5%	+0.4% (0.06)
100 passes (default 500)	−1.1%	−1.1%	+1.8% (<0.001)
2000 passes	−2.0%	−1.2%	+0.7% (0.008)

ones. The local improvement is needed: without it the heights increase by 0.4% ($p = 0.10$) and, in a separate run of all the instances, only 19 instead of 24 of the 27 optima of Section 4.4 are found, because the height of the best solution then decreases only in the discrete steps of the oscillation. The other elements have no significant effect ($p \geq 0.06$): the wall term ($\sigma = 0$: +0.5%), the tuned greedy criterion (greedy by volume: +0.3%), the step ρ, and the kick (no kick: +0.2%). On these instances the diversification thus comes from the restart itself, with reset weights and a new random order of the moves; the random move is needed only on small instances (Section 4.4).

4.3. Comparison with the literature

Table 3 compares the method with the literature. On the instances that share the exact geometry of 3DNest (Stoyan 2005 Examples 2–3 and Merged1–5), the method obtains lower heights in six of seven cases; the exception is Stoyan 2005 Example 2 (19.14 versus 19.01). The improvement grows with the size of the instance: on Merged3–5 our best heights are 7–9% below the best known values, obtained by the voxel VNS with the finest

Table 3: Heights (lower is better). Exact: 3DNest (Egeblad et al., 2009) (best and mean of 10 runs of 600 s), optimum of Tollenaere et al. (2026) for Stoyan 2005 Example 1, and Stoyan et al. (2004) (St04). Voxels: VNS of Lamas-Fernandez et al. (2023) (best: standard resolution and 10 runs of 1 h for Stoyan, extra-fine resolution and 4 h for Merged; mean: standard resolution). Ours: mean (best) of 10 seeds, 120 s. *Optimal (equal to the lower bound H_0). †Value reported by Lamas-Fernandez et al. (2023); it is below the height of the tallest piece (29) in the public data. ‡HAPE3D (Liu et al., 2015), allowing 8 rotations. §Not exact: conservative convex decomposition (Section 4, Discussion); ours: 5 seeds of 20 min (15 min with a coarse decomposition and 5 min with a fine one); voxels: 10 runs of 1 h.

		Exact		Voxels		
Instance	n	best	mean	best	mean	Ours
Stoyan 2005 Ex1	7	17.46 (opt.)	–	–	–	17.47 (17.46)
Stoyan 2005 Ex2	12	**19.01**	19.83	21.23	21.70	19.16 (19.14)
Stoyan 2005 Ex3	25	28.83	29.82	29.76	31.53	**27.45 (27.05)**
Stoyan 2004 Ex2	20	32 (St04)	–	28.47†	29.40	**29.00 (29.00)***
Stoyan 2004 Ex3	30	49 (St04)	–	41.30	42.65	**40.00 (39.13)**
Stoyan 2004 Ex4	40	63.21 (St04)	–	54.83	59.52	**53.32 (52.15)**
Stoyan 2004 Ex5	50	78.7 (St04)	–	72.33	76.81	**66.96 (65.13)**
Merged1	15	17.00	17.45	17.13	18.67	**16.70 (16.56)**
Merged2	30	21.98	22.02	22.06	22.64	**21.10 (21.01)**
Merged3	45	24.04	24.97	24.32	26.26	**22.60 (22.01)**
Merged4	60	27.81	28.35	26.47	29.44	**24.93 (24.12)**
Merged5	75	30.13	31.31	29.24	31.12	**27.93 (27.07)**
Liu Exp. 1	36	31.2‡	–	34.27	35.37	**30.22 (30.00)**
Curved meshes (conservative decomposition)						
Chess§	32	–	–	**131**	**135.1**	148.4 (138.6)

resolution and four hours, and our mean heights after 120 s are lower than those best values. On Stoyan 2004, the method improves the exact reference by a wide margin, and on Example 2 it reaches the optimum.

On Stoyan 2004 Example 2 our height, 29, equals the lower bound H_0 given by the tallest piece, so it is optimal for the public data. Stoyan 2004 Example 1 (80 pieces) has no published reference and is not included; the instance called Experiment 1 by Lamas-Fernandez et al. (2023) is the one of Liu et al. (2015), where our mean height is 12% below the best voxel height and also below that of HAPE3D, which allows rotations. For Stoyan 2005 Example 1, the utilization reported by Egeblad et al. (2009) converts to a

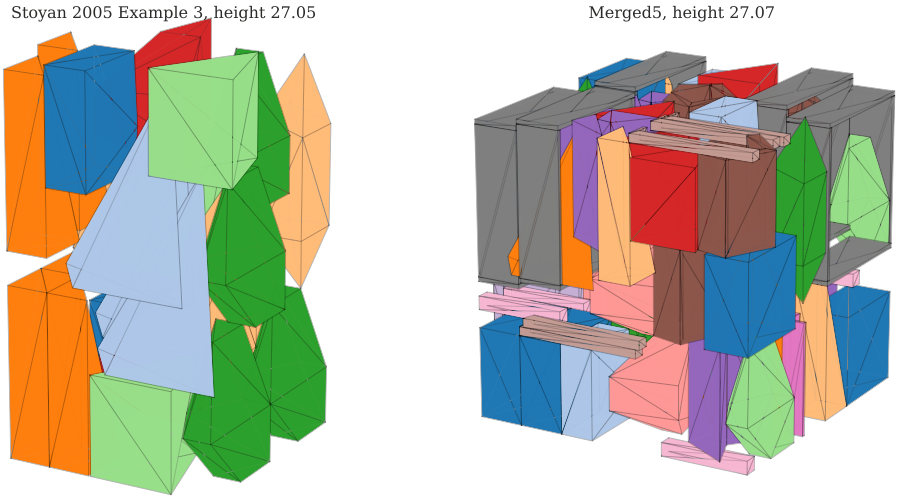


Figure 6: Best solutions of the final runs for Stoyan 2005 Example 3 and Merged5.

height below the proven optimum with our data, so the instance seems to differ and we only compare with the optimum. Figure 6 shows two of the best solutions.

4.4. Comparison with exact models

Tollenaere et al. (2026) solve four series of small instances (2–20 items of the convex pieces of Stoyan 2005 and Stoyan 2004, of diamonds and of tetrahedra) with mixed-integer models based on no-fit polyhedra, with a time limit of one hour per model; they prove optimality for 27 of the 66 instances and report upper and lower bounds for the others. Table 4 summarizes our results (10 seeds, 30 s per instance). The method finds 24 of the 27 proven optima in its best run, 12 of them in every run. Two of the others are missed by 0.03% (19.31 versus 19.30 and 24.94 versus 24.93, the 3-item instance of the Stoyan 2004 series and the 9-item instance of the Stoyan 2005 series), and the 10-item instance of the Stoyan 2005 series by 0.6% (26.20 versus 26.05). On the 39 open instances it improves the best upper bound in 32 cases, by 5% on average, and matches it in most of the others. The random move of the restarts is needed here: in an earlier version, whose restarts only reinserted the moved piece at its lowest position, the optimum of two 2-item instances was missed (by 17% and 28%), because the greedy places the larger piece on the floor, the optimum needs the opposite arrangement, and neither

Table 4: Instances of Tollenaere et al. (2026) (10 seeds, 30 s). Optima: instances with proven optimum, and number of them found in the best run and in all runs. Open: instances without proven optimum, number in which our best height is below the best upper bound of the models, and mean gap of our mean height with respect to that bound.

		Optima			Open	
Series	n	proven	best run	all runs	improved	gap
Diamonds	9	4	4	2	5 of 5	−4.5%
Stoyan 2004	19	9	8	3	9 of 10	−4.8%
Stoyan 2005	19	9	7	5	9 of 10	−4.7%
Tetrahedra	19	5	5	2	9 of 14	−1.7%
Total	66	27	24	12	32 of 39	

the GLS weights (a single pair) nor reinsertion can escape.

4.5. Time-matched comparison

Hardware differs by about ten years: 3DNest ran on one core of a 2.16 GHz Intel Core Duo (2006), and the single-thread ratings of PassMark (PassMark Software, 2026) are 667 for that processor (Core Duo T2600) and 1624 for ours (Xeon E5-2620 v4), a factor of about 2.4 per core; this is rather an upper bound, since the rating uses the maximum turbo frequency and our runs share the processor. Figure 7 compares the mean utilization over time with that of 3DNest at 10, 60, 300 and 600 seconds, converted to our machine. Our first solution is available after about 0.3 seconds (the greedy construction), and at 10 seconds our mean utilization already exceeds the mean utilization of 3DNest at 600 seconds on all five instances (44.1, 43.1, 44.9, 43.8 and 43.1% versus 43.0, 42.9, 43.3, 42.0 and 41.0%). With the factor 2.4, our 10 seconds correspond to about 24 seconds on the machine of Egeblad et al. (2009), and their 600 seconds to about 250 seconds on ours, so the advantage holds at equal effective time by more than an order of magnitude.

4.6. Discussion

The results support a conclusion that agrees with recent 2D work (Gardeyn et al., 2025): the quality of an overlap-minimization heuristic is driven by how many separation attempts fit in the time budget, and therefore by the cost of evaluating and optimizing the overlap. The NFP-based penetration depth gives an exact, cheap and informative measure for unions of convex parts with few facets, which is the case of all the benchmark instances.

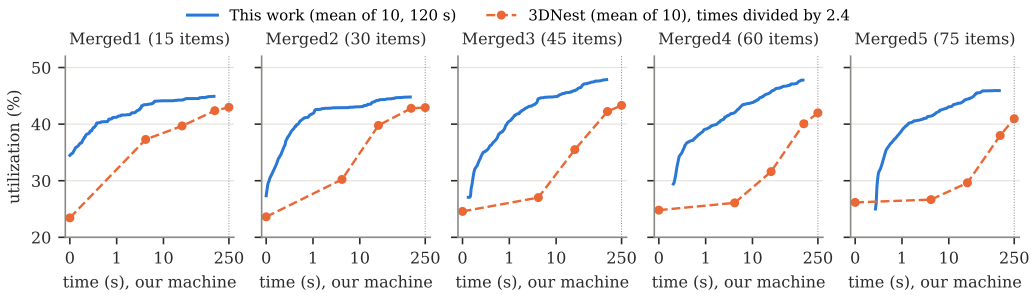


Figure 7: Mean utilization over time on the Merged instances (10 seeds), compared with 3DNest (Egeblad et al., 2009, Table 3) (10 runs). The times of 3DNest (10, 60, 300 and 600 s) are divided by 2.4, the ratio of the single-thread performance of the two processors, so that both curves are on the time scale of our machine; the dotted line marks the 600 s of 3DNest (250 s).

Realistic meshes. The cost of a move grows with the number of facets of the NFPs, so pieces with curved surfaces are the hard case. We tested the Chess instance of Lamas-Fernandez et al. (2023) (32 pieces, meshes of 700–2900 triangles). Since an exact convex decomposition of such meshes is impractical, we use a conservative one (the large convex hulls of the NFPs are computed with Qhull (Barber et al., 1996)): CoACD (Wei et al., 2022) cuts the mesh and takes the convex hull of each piece; the small gaps it leaves between parts are covered by scaling each part from its centroid, and containment is checked with interior sample points. With concavity threshold 0.05 (6–20 parts per piece) the union exceeds the volume of the piece by 4–12%, and with 0.2 (1–4 parts, reduced to at most 42 facet directions) by 38–78%. The solutions remain feasible for the real geometry. However, the NFPs between parts have hundreds of facets and a separation takes 20–35 s instead of a fraction of a second. In 20 minutes (5 seeds; Table 3) the best (mean) heights are 146.3 (151.2) with the fine decomposition, 149.9 (155.0) with the coarse one and 138.6 (148.4) with 15 minutes of coarse search followed by 5 minutes of refinement with the fine decomposition, against 131 (best of 10 runs of 1 h) and 135.1 (mean) for the voxel VNS of Lamas-Fernandez et al. (2023), whose cost does not depend on the complexity of the surface. Pseudo-blocks of two pieces did not help. The other realistic instance of Lamas-Fernandez et al. (2023), Engine (97 pieces with meshes of up to 342,000 triangles), was not tested, for the same reason. Voxels and the proposed method are thus complementary: the latter is faster and exact on polyhedral pieces with few

facets, the former scales better to curved meshes.

Limitations. Rotations are not considered. The method requires a convex decomposition of each piece; for the benchmark instances it is exact and small, while for curved meshes a conservative decomposition overestimates the pieces and, above all, makes the moves expensive. Reducing the cost of the NFPs of curved parts (e.g., hierarchical filtering with the hull of each piece, or conservative simplification of the NFPs) is left for future work.

5. Conclusions

We presented an overlap-minimization heuristic for 3D irregular strip packing that works with the exact geometry of polyhedra given as unions of convex parts. Its key element is an exact axis move based on penetration depths in no-fit polytopes, combined with multiplicative guided local search, strategic oscillation and restarts. This simple method outperforms the previous methods, with exact and voxel geometry, on the benchmarks of polyhedral pieces, and matches almost all the proven optima of the small instances. On curved meshes, which require conservative decompositions with many facets, it remains behind voxel-based search. Future work includes cheaper NFP computations for curved parts, rotations, larger instances from additive manufacturing, and lower bounds.

Acknowledgments

This work was supported by ANID FONDECYT project 1260159.

CRediT authorship contribution statement

Ignacio Araya: Conceptualization, Methodology, Software, Validation, Investigation, Writing – original draft, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The source code, the convex decompositions of the pieces, the best solutions of Table 3 and the scripts that reproduce all the experiments are available at `https://github.com/rilianx/poly3d-packing`. The instances are available in the ESICUP repository (EURO Special Interest Group on Cutting and Packing, 2025).

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